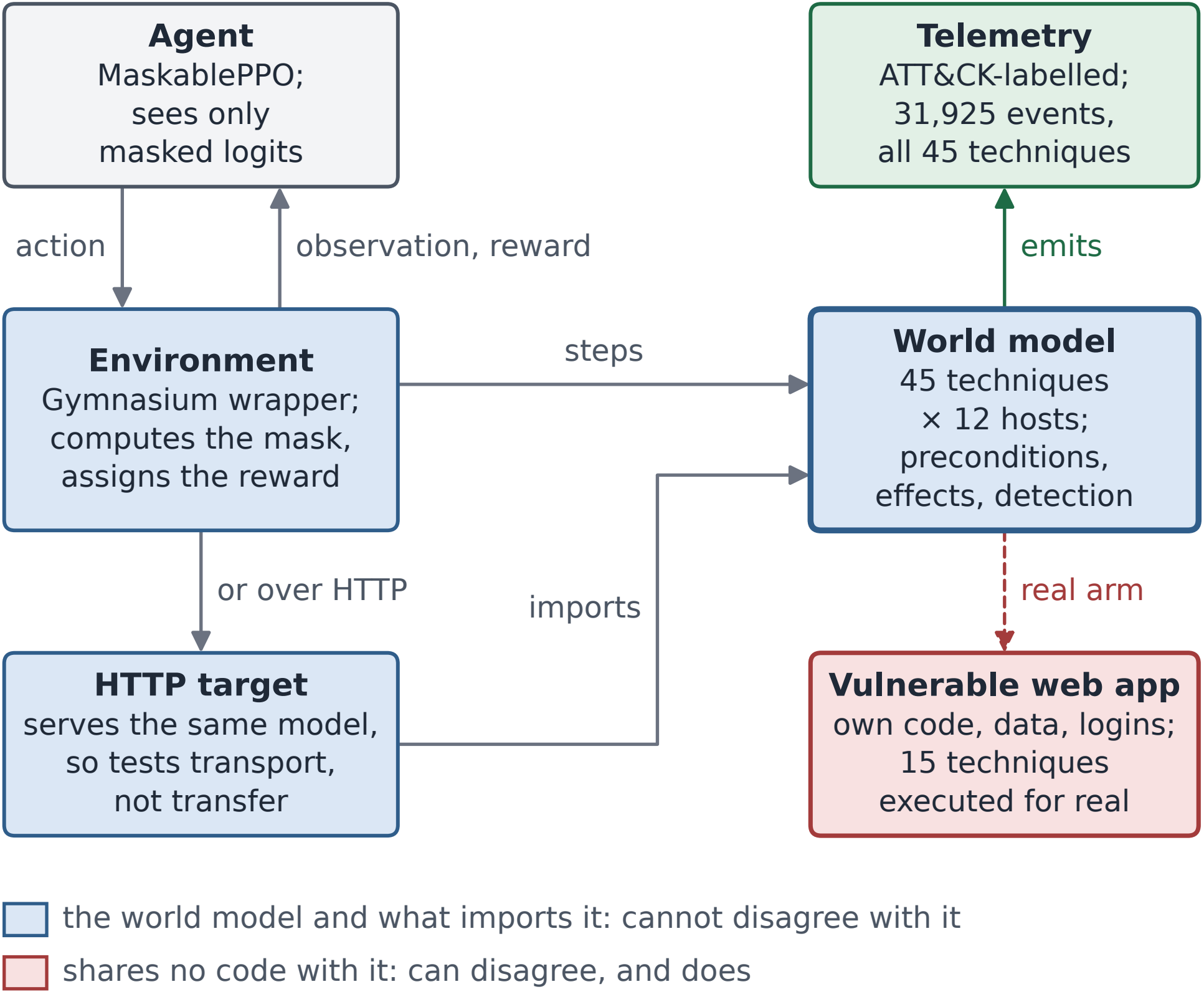
Agent
MaskablePPO;
sees only
masked logits
Telemetry
ATT&CK-labelled;
31,925 events,
all 45 techniques
action
observation, reward
emits
Environment
Gymnasium wrapper;
computes the mask,
assigns the reward
steps
World model
45 techniques
× 12 hosts;
preconditions,
effects, detection
or over HTTP
imports
real arm
HTTP target
serves the same model,
so tests transport,
not transfer
Vulnerable web app
own code, data, logins;
15 techniques
executed for real
the world model and what imports it: cannot disagree with it
shares no code with it: can disagree, and does